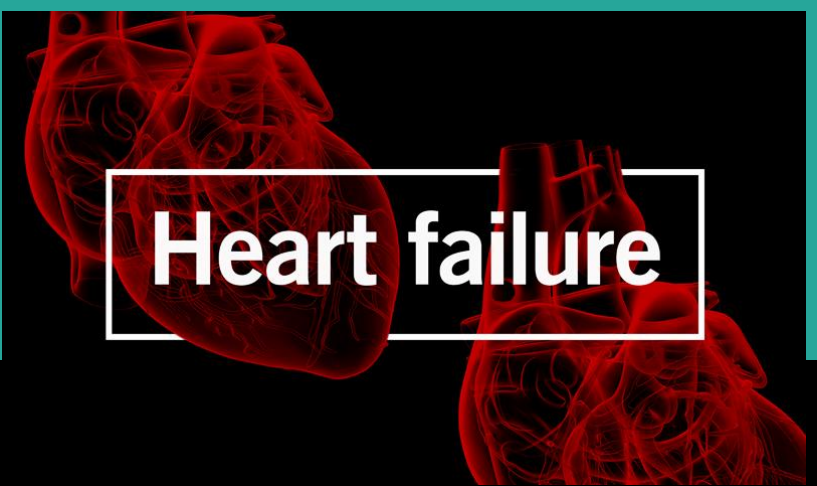




Heart failure

Heart Failure Prediction

(using Linear and Logistic Regression models)

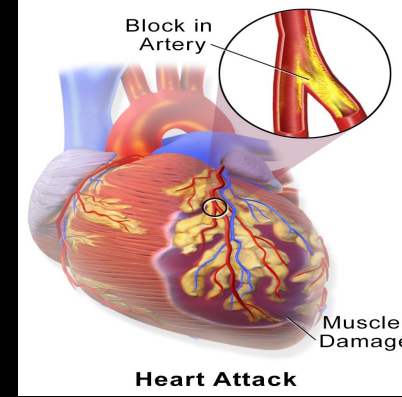
- Soham Sarkar (Course: STAT 5353)

Heart Failure

Heart failure (HF) is a type of cardiovascular diseases (CVD) in which, the heart is unable to pump blood to the rest of the body, and is one of the primary heart disorders which affect 17 million people worldwide each year along with heart attacks and strokes.

Clinical community groups HF into two categories based on how the left ventricle is able to pump out blood through the blood vessels.

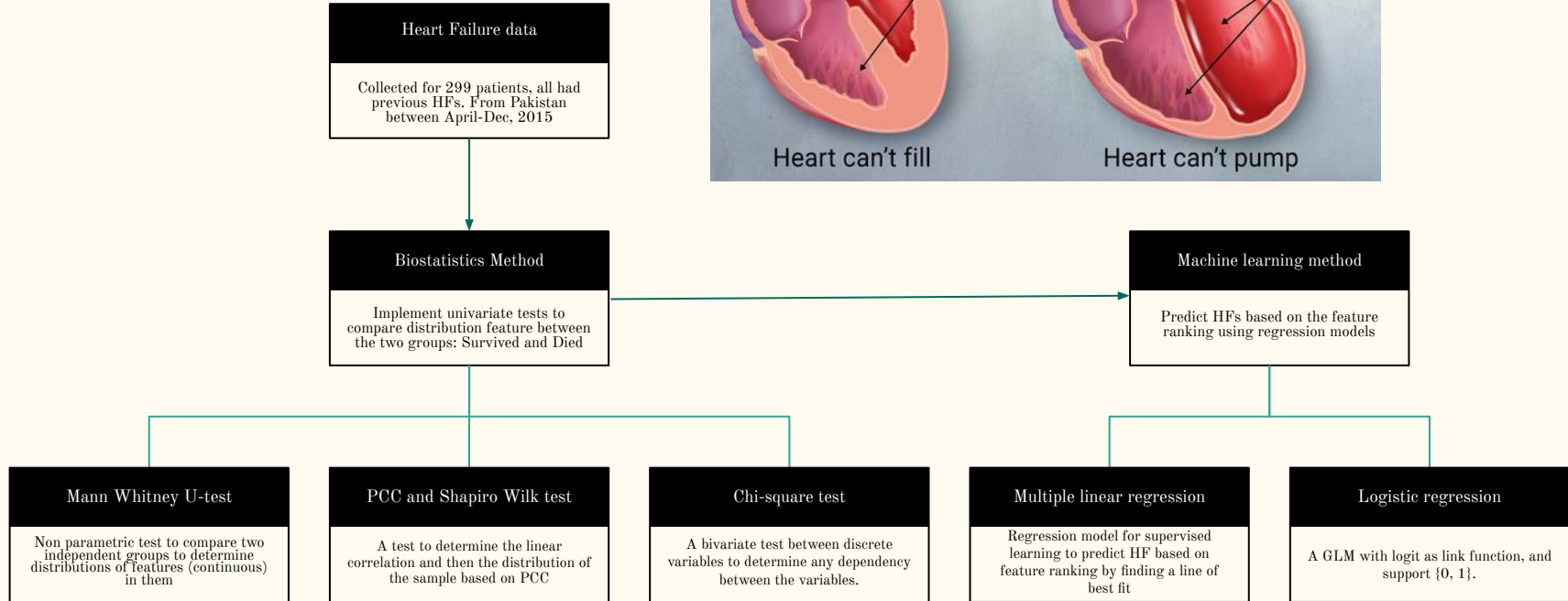
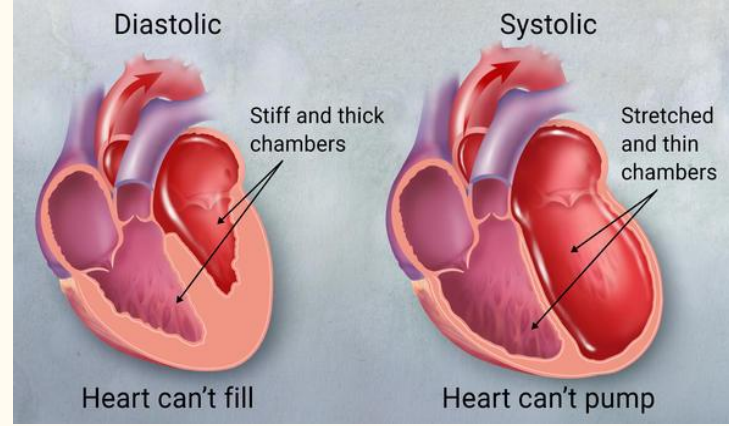
This data is collected for 299 patients with 13 risk factors in 2015 all of whom have previously been diagnosed with HF, and have been admitted at hospitals in Faisalabad, Pakistan.



Goal:

- Rank clinically important risk factors from HF dataset, which is difficult to predict by formal testing
 - To implement tools from ML for quantifying this observation and forecasting HF
 - Compare results of the ML models in predictions
-

Outline of the approach



Dataset

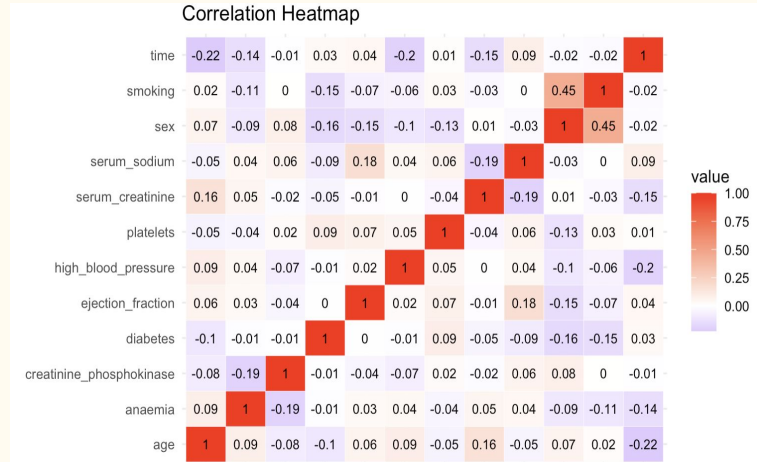
Feature	Explanation	Measurement	Range
Age	Age of the patient	Years	[40,..., 95]
Anaemia	Decrease of red blood cells or hemoglobin	Boolean	0, 1
High blood pressure	If a patient has hypertension	Boolean	0, 1
Creatinine phosphokinase (CPK)	Level of the CPK enzyme in the blood	mcg/L	[23,..., 7861]
Diabetes	If the patient has diabetes	Boolean	0, 1
Ejection fraction	Percentage of blood leaving the heart at each contraction	Percentage	[14,..., 80]
Sex	Woman or man	Binary	0, 1
Platelets	Platelets in the blood	kiloplatelets/mL	[25.01,..., 850.00]
Serum creatinine	Level of creatinine in the blood	mg/dL	[0.50,..., 9.40]
Serum sodium	Level of sodium in the blood	mEq/L	[114,..., 148]
Smoking	If the patient smokes	Boolean	0, 1
Time	Follow-up period	Days	[4,...,285]
(target) death event	If the patient died during the follow-up period	Boolean	0, 1

Feature overview

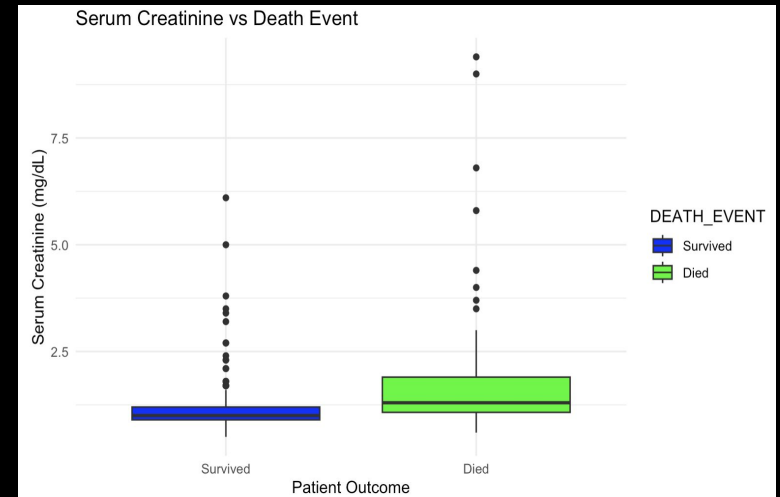
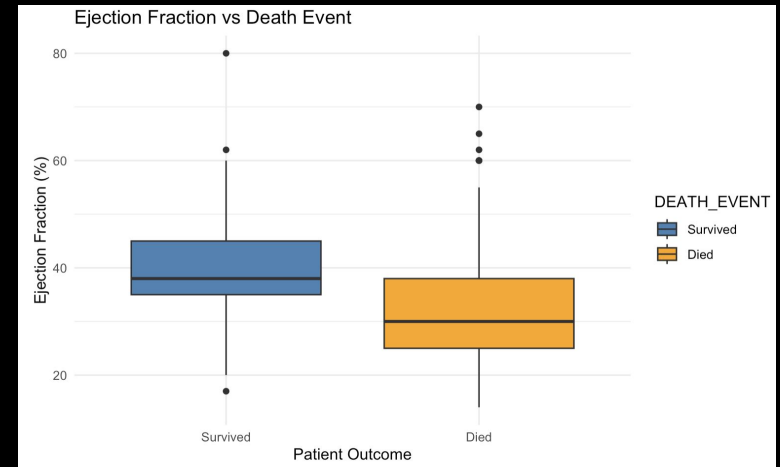
- The 13 features include clinical, body and lifestyle informations.
- All the patients have been diagnosed previously with left ventricular systolic dysfunction.
- The dataset has 203 patients who survived and 96 patients who died.
- 194 men and 105 women aged between 40 and 95 years.
- Follow up period is the time after the initial diagnosed HF.

We perform a binary classification study that the patient admitted has died (death event = 1) or survived(death event = 0) before the end of the follow-up period.

Feature relations



- Correlation heatmap determines which features have positive and negative correlations between them
- Boxplot of Ejection fraction vs Death event implies higher survival rate for more ejection fraction
- Boxplot for serum creatinine vs Death event implies lower levels of serum creatinine implies higher survival rate



Biostatistics method: Feature ranking method

Mann Whitney U-test

Used to detect if we can reject the null hypothesis that the distribution of each feature for death event, are the same.

Here, a p-value lower than 0.05 implies that the analyzed feature strongly relates to death event, whereas the p-value higher than 0.05 means the opposite.

In our HF dataset, we see two features having p-value very close to 0, which are 'serum creatinine' and 'ejection fraction'.

PCC and Shapiro-Wilk test

The Pearson correlation coefficient (PCC) indicates linear correlation between features from the two groups.

If the absolute value of PCC is high (close to 1), then the elements have a linear correlation, and a low value (close to 0) implies otherwise.

In HF dataset, 'age' is also an important risk factor. Since the Shapiro-Wilk test generates p-value close to 0 (and less than 0.05) for all features, the null hypothesis of normality is rejected. So, all variables are non-normally distributed.

Chi - square test

It is used to determine if two discrete features are independent.

So, a low p-value indicates a strong dependency between the two variables, whereas a high p-value indicates that the null hypothesis of independence between the two features can not be rejected.

Since in our dataset, we observed that the p-value for the features 'serum creatinine' and 'ejection fraction' are less than 0.05, they are strongly related.

Feature ranking results

Mann-Whitney U		
Rank	Feature	Test p-value
1	Serum creatinine	0
2	Ejection fraction	0.000001
3	Age	0.000167
4	Serum sodium	0.000293
5	High blood pressure	0.171016
6	Anaemia	0.252970
7	Platelets	0.425559
8	Creatinine phosphokinase	0.684040
9	Smoking	0.828190
10	Sex	0.941292
11	Diabetes	0.973913

- Mann-Whitney test indicates the distribution of ‘serum creatinine’ and ‘ejection fraction’ for the death event to be different.
- PCC and Shapiro-Wilk test indicates non-normal distribution for all the features.
- The chi-square test proves that the top 2 features have a strong chance of dependency.

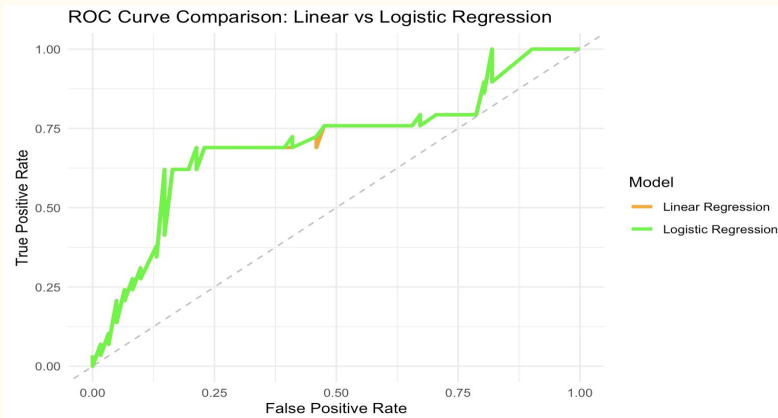
Pearson correlation coefficient			Shapiro-Wilk test		
Rank	Feature	abs(PCC)	Rank	Feature	p-value
1	Serum creatinine	0.294	1	Creatinine phosphokinase	7.05×10^{-28}
2	Ejection fraction	0.269	2	Serum creatinine	5.39×10^{-27}
3	Age	0.254	3	Smoking	4.58×10^{-26}
4	Serum sodium	0.195	4	Death event	4.58×10^{-26}
5	High blood pressure	0.079	5	Sex	1.17×10^{-25}
6	Anaemia	0.066	6	High blood pressure	1.17×10^{-25}
7	Creatinine phosphokinase	0.063	7	Diabetes	5.12×10^{-25}
8	Platelets	0.049	8	Anaemia	6.21×10^{-25}
9	Smoking	0.013	9	Platelets	2.89×10^{-12}
10	Sex	0.004	10	Serum sodium	9.21×10^{-10}
11	Diabetes	0.002	11	Ejection fraction	7.22×10^{-09}
			12	Age	5.34×10^{-05}

Chi squared test		
Rank	Feature	p-value
1	Ejection fraction	0.000500
2	Serum creatinine	0.000500
3	Serum sodium	0.003998
4	Age	0.005997
5	High blood pressure	0.181909
6	Anaemia	0.260370
7	Creatinine phosphokinase	0.377811
8	Platelets	0.637681
9	Smoking	0.889555
10	Sex	1
11	Diabetes	1

Machine learning models

We use multiple linear regression and logistic regression models and compare the predictions for each of them using confusion matrix metrics (MCC, ROC (AUC), F1 score and accuracy).

```
Actual
Predicted 0 1
          0 57 23
          1  4  6
Accuracy: 0.7
Setting levels: control = 0, case = 1
Setting direction: controls < cases
AUC: 0.7041
F1 Score: 0.3077
Matthews Correlation Coefficient (MCC): 0.2102
```



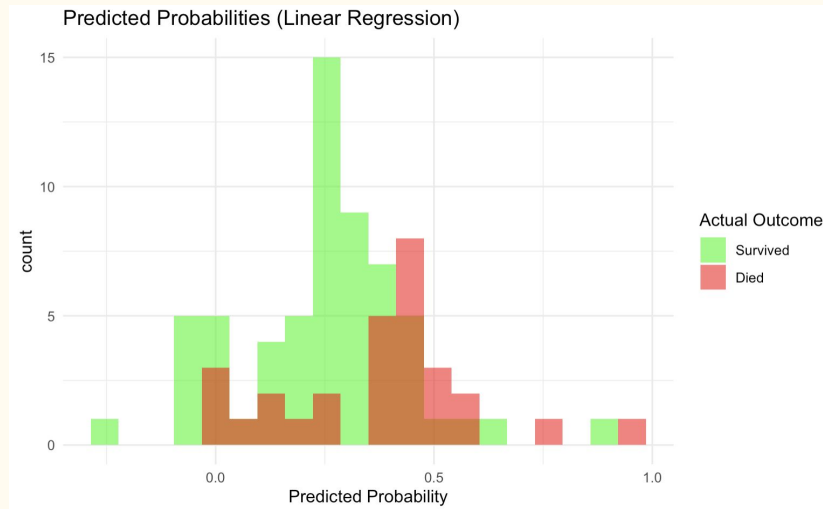
```
Actual
Predicted 0 1
          0 56 21
          1  5  8
Accuracy: 0.7111
Setting levels: control = 0, case = 1
Setting direction: controls < cases
AUC: 0.7058
F1 Score: 0.381
Matthews Correlation Coefficient (MCC): 0.2578
```

From the confusion matrix, the total of the number of TPs and TNs is higher for logistic regression. The accuracy, AUC and F1 score are higher in comparison to the multiple linear regression model applied.

We evaluate MCC, which lies between -1 and 1, and is a more balanced evaluator, since it considers all 4 quadrants of the confusion matrix.

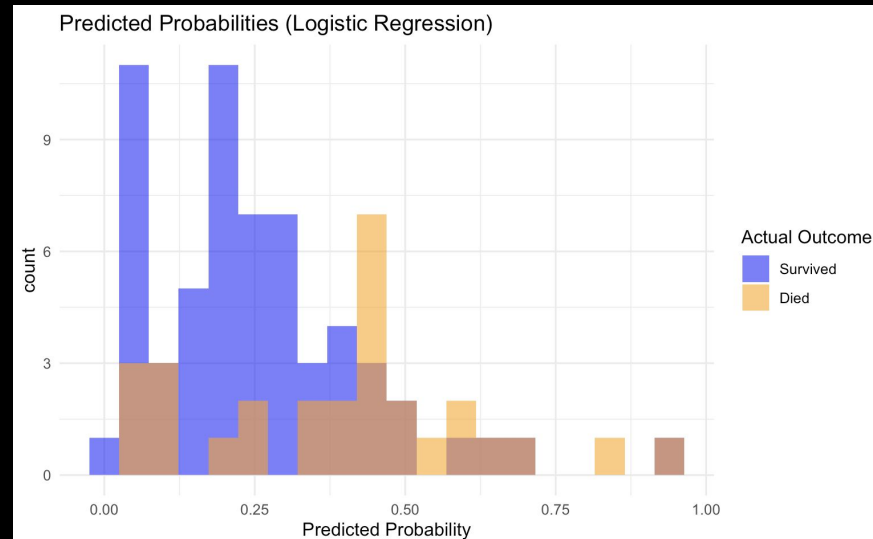
The higher MCC value for logistic regression model implies stronger positive correlation between the model's predictions and the actual outcome suggesting better performance than the linear regression model.

Prediction



- More histograms around 0.5 implies model is uncertain in predictions, true for linear regression model.
- There is some overlap for both models and so, both models mispredicted in those cases.

Logistic regression is more confident in predicting death event than linear regression.



Conclusion

We have been able to select the two most relevant features to predict HF, and concluded that ML algorithms can be used for binary classification from electronic health records of patients with CVDs.

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- 4) Wilcoxon F. Individual comparisons by ranking methods. Biom Bull. 1945; 1(6):80–3.

Thank You for listening
